

WRF hindcast report

Reproducible mesoscale hindcast and wind-resource screening

Parameter	Result
Simulation period	2025-01-10T00:00:00 to 2025-01-10T06:00:00 UTC
Domains	2
Primary analysis domain	d02
Outputs per domain	7
Report scope	Meteorology, surface fluxes, precipitation, PBL and 10 m wind-resource screening

Interpretation boundary

This technical report verifies model output and summarizes the simulated period. It is not a climatology, validation against observations, bankable wind-resource assessment, or substitute for physics sensitivity testing.

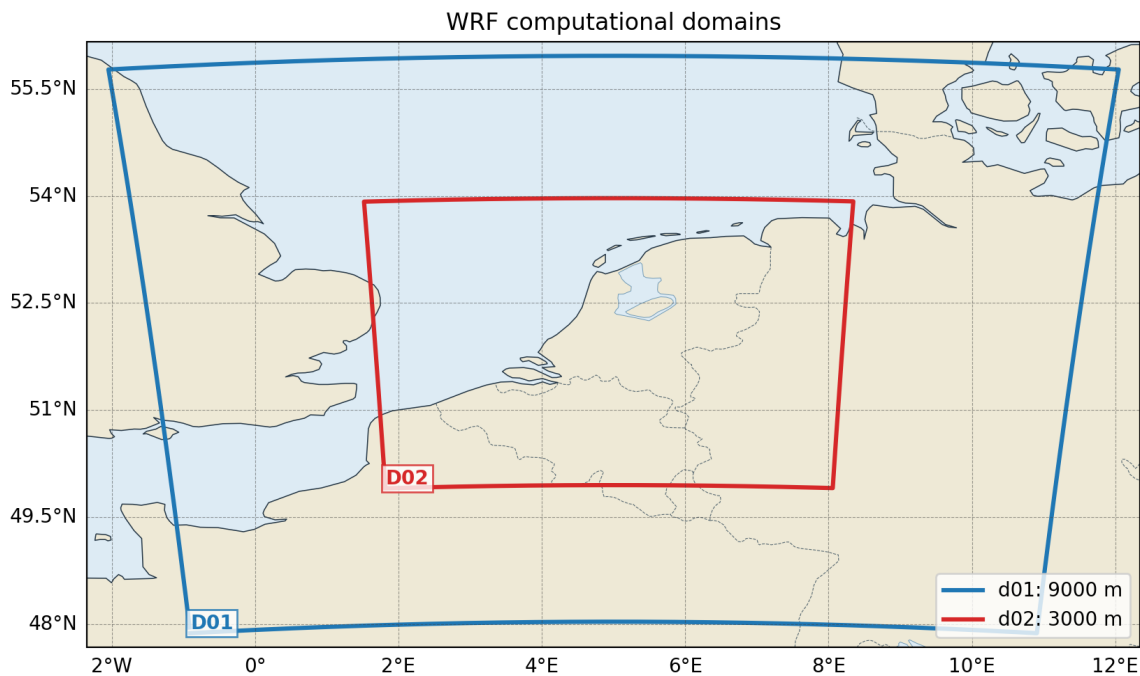


Figure 1. Computational-domain boundaries and nominal grid spacing.

Domain configuration comparison

Temporal configuration

Parameter	Result
ERA5 forcing interval	3600 seconds (hourly)
WRF output interval	60 minutes
Integration timestep	D01: 54 seconds, D02: 18 seconds
Restart checkpoint interval	180 minutes
Resumed from	Fresh initialization

Grid

Parameter	D01	D02
Grid ID	1	2
Horizontal grid	99 x 99	150 x 150
Spacing	9000 x 9000 m	3000 x 3000 m
Vertical levels	39	39
Output times	7	7

Physics suite

Parameter	D01	D02
MP PHYSICS	4	4
RA LW PHYSICS	4	4
RA SW PHYSICS	4	4
SF SFCLAY PHYSICS	1	1
SF SURFACE PHYSICS	2	2
BL PBL PHYSICS	1	1
CU PHYSICS	1	1

Domain results comparison

Wind-resource screening

Parameter	D01	D02
Diagnostic height (m)	10.00	10.00
Mean wind speed (m s-1)	5.76	6.73
Maximum wind speed (m s-1)	13.09	13.09
P50 wind speed (m s-1)	5.22	6.04
P90 wind speed (m s-1)	10.38	11.43
P95 wind speed (m s-1)	11.36	12.14
Mean direction (degrees)	271.38	270.19
Mean air density (kg m-3)	1.27	1.28
Mean wind-power density (W m-2)	226.96	311.46

Vertical-profile diagnostics

Parameter	D01	D02
Shear fit layer	40.0-200.0 m	40.0-200.0 m
Mean shear exponent	0.295	0.311
TKE diagnostic	not present for this physics/output configuration (no variable)	not present for this physics/output configuration (no variable)

Meteorological means and ranges

Parameter	D01	D02
T2	275 K (min 262; max 281)	275 K (min 266; max 280)
PSFC	1e+05 Pa (min 9.06e+04; max 1.02e+05)	1.01e+05 Pa (min 9.43e+04; max 1.02e+05)
Q2	0.00365 kg kg-1 (min 0.00154; max 0.00516)	0.00368 kg kg-1 (min 0.00217; max 0.00481)
U10	4.11 m s-1 (min -6.19; max 11.5)	5.87 m s-1 (min -2.01; max 11.3)
V10	-1.01 m s-1 (min -11.1; max 6.48)	-1.01 m s-1 (min -9.19; max 5.22)
PBLH	414 m (min 0; max 1.28e+03)	443 m (min 0; max 1.4e+03)
RAINC	0.0195 mm (min 0; max 2.86)	0.0094 mm (min 0; max 2.2)
RAINNC	0.033 mm (min 0; max 4.06)	0.0288 mm (min 0; max 2.91)
HFX	0.994 W m-2 (min -77.5; max 98.5)	0.845 W m-2 (min -93.2; max 99.2)
LH	21.7 W m-2 (min -23.4; max 136)	26.8 W m-2 (min -24; max 136)
SWDOWN	0 W m-2 (min 0; max 0)	0 W m-2 (min 0; max 0)
GLW	210 W m-2 (min 0; max 333)	211 W m-2 (min 0; max 327)
UST	0.282 m s-1 (min 0.0001; max 1.12)	0.329 m s-1 (min 0.0001; max 1.03)

Quality-control overview

Parameter	D01	D02
QC status	ISSUES_FOUND	ISSUES_FOUND
Errors	0	0
Warnings	5	6

D01 QC details

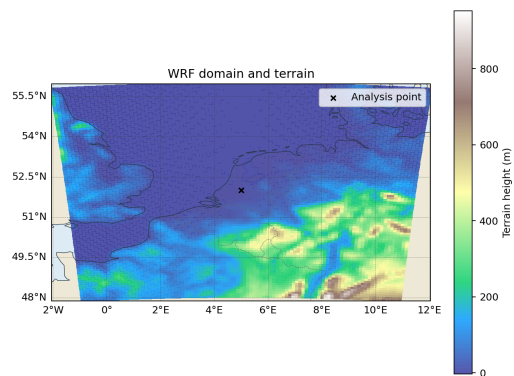
Parameter	Result
WARNING - T2	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - PBLH	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - LH	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - GLW	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - UST	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1

D02 QC details

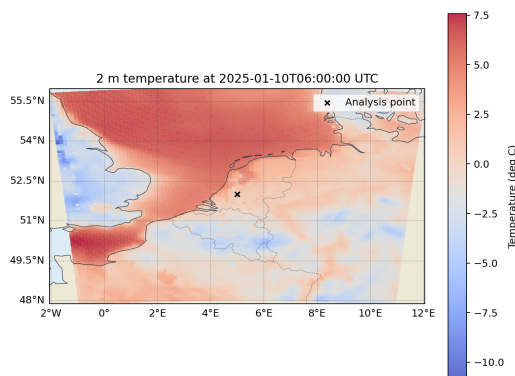
Parameter	Result
WARNING - T2	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - PBLH	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - RAINC	2 domain-mean jumps exceed 12 robust standard deviations; first at Time=5
WARNING - LH	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - GLW	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1
WARNING - UST	1 domain-mean jumps exceed 12 robust standard deviations; first at Time=1

Domain d01 figures

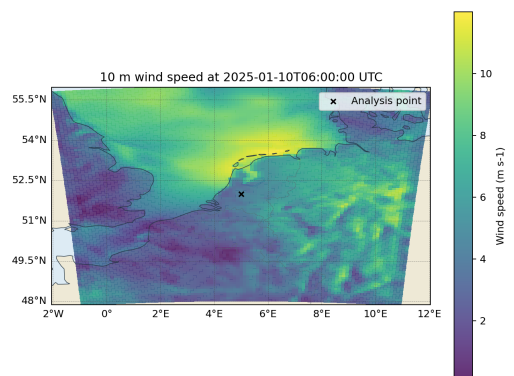
Grid ID 1 | 9000 m horizontal spacing



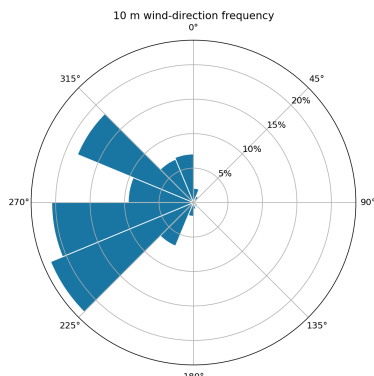
D01 Figure 1. Domain Terrain.



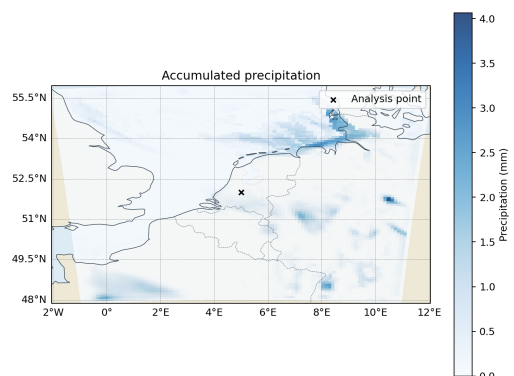
D01 Figure 2. Final Temperature.



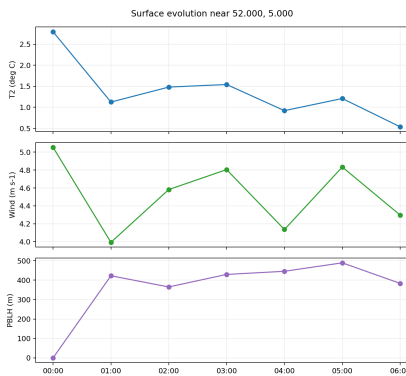
D01 Figure 3. Final Wind Speed.



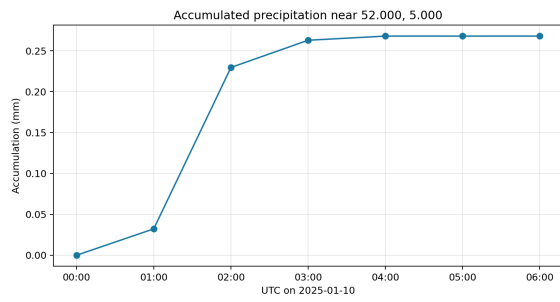
D01 Figure 4. Wind Rose.



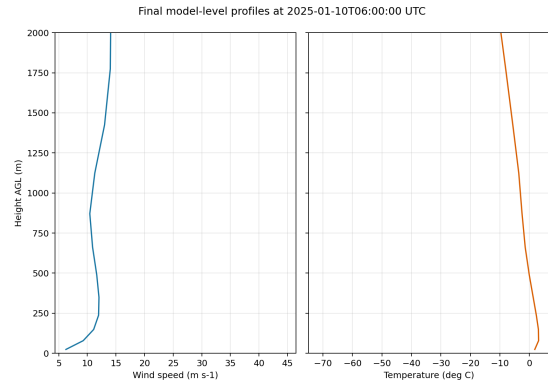
D01 Figure 5. Accumulated Precipitation.



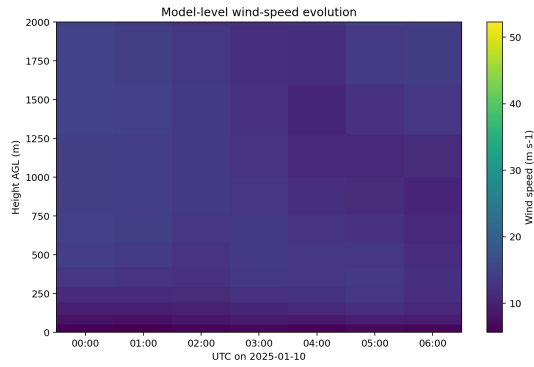
D01 Figure 6. Selected Point Timeseries.



D01 Figure 7. Precipitation Timeseries.



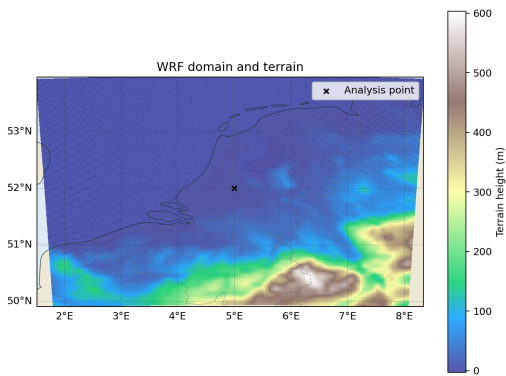
D01 Figure 8. Vertical Profiles.



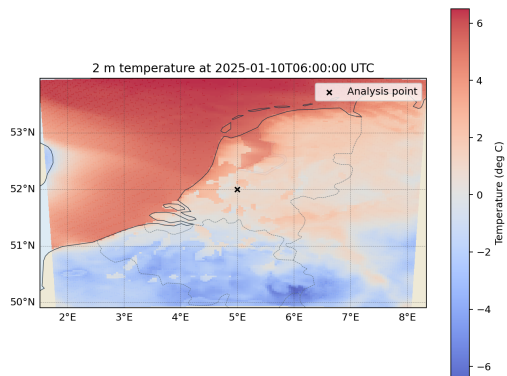
D01 Figure 9. Wind Time Height.

Domain d02 figures

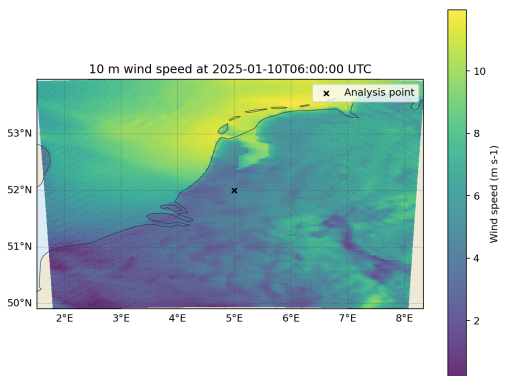
Grid ID 2 | 3000 m horizontal spacing



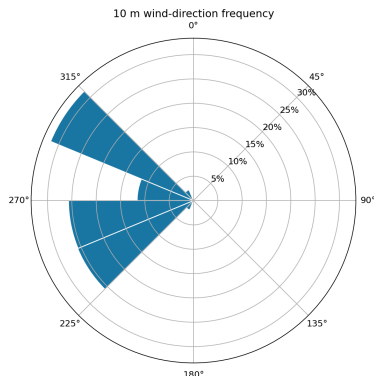
D02 Figure 1. Domain Terrain.



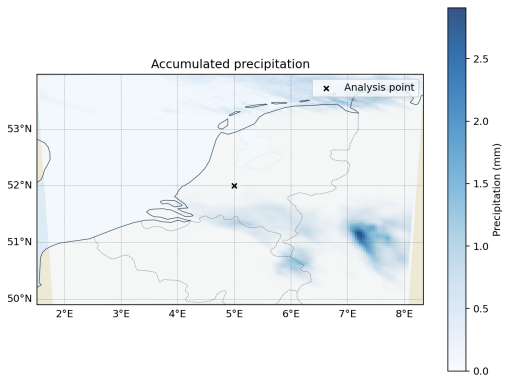
D02 Figure 2. Final Temperature.



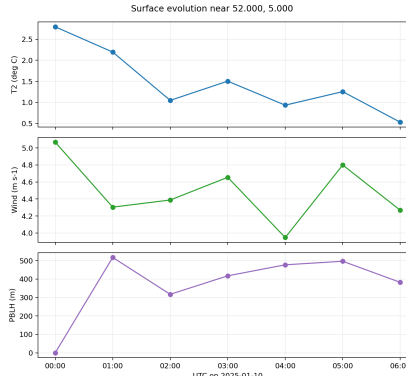
D02 Figure 3. Final Wind Speed.



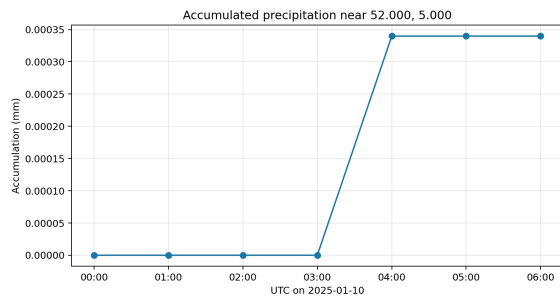
D02 Figure 4. Wind Rose.



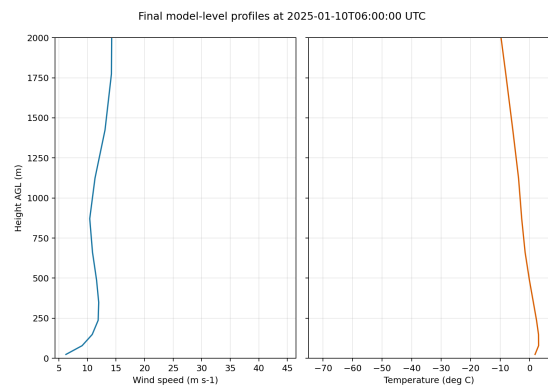
D02 Figure 5. Accumulated Precipitation.



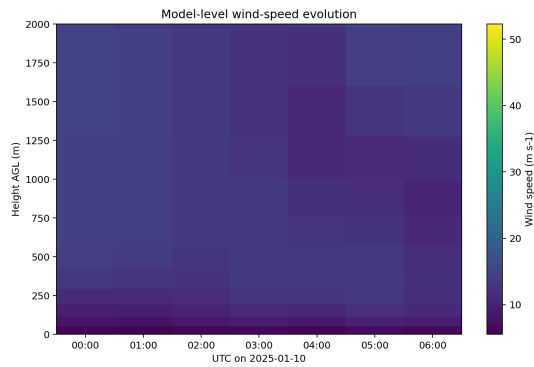
D02 Figure 6. Selected Point Timeseries.



D02 Figure 7. Precipitation Timeseries.



D02 Figure 8. Vertical Profiles.



D02 Figure 9. Wind Time Height.

Reproducibility and provenance

Parameter	Result
inputs	["wrfout_d01_2025-01-10_00:00:00", "wrfout_d02_2025-01-10_00:00:00", "wrfout_d01_2025-01-10_01:00:00", "wrfout_d02_2025-01-10_01:00:00", "wrfout_d01_2025-01-10_02:00:00", "wrfout_d02_2025-01-10_02:00:00", "wrfout_d01_2025-01-10_03:00:00", "wrfout_d02_2025-01-10_03:00:00", "wrfout_d01_2025-01-10_04:00:00", "wrfout_d02_2025-01-10_04:00:00", "wrfout_d01_2025-01-10_05:00:00", "wrfout_d02_2025-01-10_05:00:00", "wrfout_d01_2025-01-10_06:00:00", "wrfout_d02_2025-01-10_06:00:00"]
Case Configuration	netherlands-nested.toml SHA-256 48cdebae6f86817078acb0f65ea7f2aac150bc974993715b7d1f6a48b7ee4047
Case Manifest	case-manifest.json SHA-256 8f4d538388ff594b8fad7b80d2f1265f32c6c547915163517d805d7a7e1372f5
Build Manifest	build-manifest.txt SHA-256 7cdcb7189f5400e5aa3a2d9cdf25039cad48c6352ac9a9b35d7d85931de6e1f2
Workflow State	workflow-state.json SHA-256 045a862b6b4d519b7676fc784a7e0b398f7ea2c37fc2db474b54adee9f1d3470
software_version	"wrf-tools 0.3.0"